$$\Rightarrow \quad y = \frac{x}{10x-3}$$

$$\therefore \quad 10xy - 3y = x$$

$$\Rightarrow \quad x + 3y = 10xy$$

59. Option (b) is correct.

First we need to find slope of the given line:

$$x\cos\theta + y\sin\theta = 9$$

Rewrite the equation as follows:

$$y\sin\theta = 9 - x\cos\theta$$

$$\Rightarrow \quad y = \frac{9}{\sin\theta} - \frac{x\cos\theta}{\sin\theta}$$

$$\Rightarrow \quad y = -\frac{x\cos\theta}{\sin\theta} + \frac{9}{\sin\theta} \quad \text{...(i)}$$

$\because$ The general equation of line is:

$$y = mx + c \quad \text{...(ii)}$$

Where (x, y) is the general point on the line. m is the slope of the line, c is the y-intercept.

On comparing equation (i) and (ii), we get

$$m = -\frac{\cos\theta}{\sin\theta}$$

We know that, the slope of perpendicular line are negative inverse of each other.

$\therefore$ The slope m_1 of the required line can be

$$m_1 = -\left(\frac{1}{-\frac{\cos\theta}{\sin\theta}}\right)$$

$$\Rightarrow \quad m_1 = \frac{\sin\theta}{\cos\theta}$$

Also, the line passes through the point $(-\sin\theta, \cos\theta)$. We know that, one point slope from the line is:

$$y - y_1 = m(x - x_1)$$

Where m is the slope and (x_1, y_1) is the given point on the line.

$\therefore$ The equation of the required line is

$$y - \cos\theta = \frac{\sin\theta}{\cos\theta}(x + \sin\theta)$$

$$\Rightarrow \quad \cos\theta\, y - \cos^2\theta = x\sin\theta + \sin^2\theta$$

$$\Rightarrow x\sin\theta - y\cos\theta + \sin^2\theta + \cos^2\theta = 0$$

$$\Rightarrow x\sin\theta - y\cos\theta + 1 = 0$$

60. Option (a) is correct.

We have given P and Q lie on line $y = 2x + 3$

For P, Put $x = a$ then $y = 2a + 3$

For Q, Put $x = b$ then $y = 2b + 3$

Coordinates $P = (a, 2a + 3)$ and $Q = (b, 2b + 3)$

$R = (1, 5)$

From using distance formula, we have

$$PR = \sqrt{(a-1)^2 + (2a+3-5)^2}$$

$$= 2 \quad \text{...(i)}$$

$$QR = \sqrt{(b-1)^2 + (2b+3-5)^2}$$

$$= 2 \quad \text{...(ii)}$$

From (i), we get

$$(a-1)^2 + (2a-2)^2 = 4$$

$$a^2 - 2a + 1 + 4a^2 + 4 - 8a = 4$$

$$5a^2 - 10a + 1 = 0$$

From (ii), we get

$$5b^2 - 10b + 1 = 0$$

By using quadratic formula

$$a = \frac{10 \pm \sqrt{100-20}}{10}$$

$$\Rightarrow \quad a = 1 \pm \frac{2\sqrt{5}}{5}$$

$$\Rightarrow \quad a = 1 \pm \frac{2}{\sqrt{5}}$$

$$\therefore \quad b = 1 \pm \frac{2}{\sqrt{5}}$$

When $a = 1 + \frac{2}{\sqrt{5}}$ then

$$\text{Coordinate of } P = (a, 2a+3)$$

$$= \left(1 + \frac{2}{\sqrt{5}}, 2 + \frac{4}{\sqrt{5}} + 3\right)$$

$$= \left(1 + \frac{2}{\sqrt{5}}, 5 + \frac{4}{\sqrt{5}}\right)$$

When $a = 1 - \frac{2}{\sqrt{5}}$ then

$$\text{Coordinate of } P = (a, 2a+3)$$

$$= \left(1 - \frac{2}{\sqrt{5}}, 2 - \frac{4}{\sqrt{5}} + 3\right)$$

$$= \left(1 - \frac{2}{\sqrt{5}}, 5 - \frac{4}{\sqrt{5}}\right)$$

Similarly coordinate of $Q = \left(1 - \frac{2}{\sqrt{5}}, 5 - \frac{4}{\sqrt{5}}\right)$ or

$$= \left(1 + \frac{2}{\sqrt{5}}, 5 + \frac{4}{\sqrt{5}}\right)$$

So we get coordinate of the point P and Q is $\left(1 + \frac{2}{\sqrt{5}}, 5 + \frac{4}{\sqrt{5}}\right), \left(1 - \frac{2}{\sqrt{5}}, 5 - \frac{4}{\sqrt{5}}\right)$